

Notes & Meeting Minutes

Sustainable Apartments Pilot – Southport, 325 Anketell St Greenway

16 June 2025

Project name	ACT Gov. Sustainable Apartments Pilot	From	GHD
Subject	Southport OC Meeting #1	Tel / email address	sustainablebusiness@act.gov.au
Date / Time	19/05/2025 6pm	Project no.	12656728
Attendees inc. company	Anna Turnbull (ACT Gov) Jacinta Evans (ACT Gov) Adrian Piani (GHD) Gerry Hackett (GHD) Chathurika Peiris (GHD) Members of the EC, Residents and Strata Managers	Location	MS Teams

Notes & Minutes

1. Introduction

- The Sustainable Apartments Pilot is carrying out electrification feasibility studies for a representative sample of multi-residential buildings in ACT, to build knowledge on the technical challenges to electrification and develop potential solutions and costings. The Pilot aims to gather learnings that will inform community on how electrification works can be undertaken in similar buildings as ACT transitions towards zero emissions by 2045.
- Southport was one of 7 Apartment complexes selected to take part in the Pilot. Sites were selected to cover a range of building types, sizes, ages, types of gas-dependent systems. Additionally, sites that posed challenges to electrification were also preferred.
- Key reasons for Southport's selection in the Pilot:
 - Included a diverse range of gas equipment present (i.e. hot water, pool heating, laundry).
 - Mixed-use (residential and commercial).
 - Represented a large apartment complex, in terms of building height, number of dwellings, etc.
 - Geographically represented the Tuggeranong region.
- The Program is being led by ACT Government's Environment Planning and Sustainable Development Directorate (EPSDD), with assistance from GHD in the role as Technical Consultants.
- The Pilot is currently in the data collection stage. This stage consists of:
 - Initial site visit, conducted on 12/5/2025.
 - As-built drawings and documentation review.
 - Owner's Corporation meeting, conducted on 19/5/2025, providing residents, OC members and Strata Managers the opportunity to provide input or feedback that will feed into the electrification feasibility studies.
- These notes summarise the findings from the site visit and discussion from the OC meeting. They capture:
 - An overview of the buildings and it's services.
 - Constraints and opportunities to electrification.
 - Requirements and opportunities.
 - Potential electrification options for consideration.

This document is in draft form. The contents, including any opinions, conclusions or recommendations contained in, or which may be implied from, this draft document must not be relied upon. GHD reserves the right, at any time, without notice, to modify or retract any part or all of the draft document. To the maximum extent permitted by law, GHD disclaims any responsibility or liability arising from or in connection with this draft document.

Notes & Minutes

2. Observations from site visit and as-built documentation review

Site overview:

- Building constructed in 2017.
- Two multi-storey residential buildings (Stage 1 East, and Stage 2 West), with common areas and basement parking:
 - 1, 2 and 3-bedroom standard apartments, in addition to walk-up terrace style apartments.
 - 2x pools: lap pool and plunge pool with pool equipment room located adjacent
 - 2x BBQ areas (electric)
 - Gym with sauna
- 3 commercial tenancies on ground floor:
 - Restaurant – all electric cooking appliances.
 - Laundromat
 - Massage therapy
- Common surface laneway with 2 loading zone and 5 parking bays.
- Heating and cooling: Split system air conditioning units per apartment.
- Hot water: 2x central systems, 1 serving each building, located on rooftops. Hot water is metered in each apartment to apportion and bill the gas consumption associated with hot water usage.
- Cooking: electric cooktops.
- Gas meters enclosure in laneway with pipes routed through basement:
 - 2x meters, one each for each building's hot water system.
 - 1x meter to pool water heating.
 - 1x meter to commercial laundromat
 - 2x spare spaces with no gas meter, piped to remaining retail tenancies for future connections.
- Power supply to site from a dedicated chamber substation on Oakden St side, with basement main switchroom.
- 50 kW photovoltaic system on the east building.
- No electric vehicle charging provisions at present.

Summary of gas equipment, for inclusion in the electrification feasibility study:

- Hot water systems: 2x central hot water systems, each consisting of banks of gas instantaneous hot water units (14x for East building, 12x for West building).
 - During the site visit 1 unit for East building's hot water system was not functional.
 - Building Manager reported there is generally little to no perceived difference in performance from occupants when 1 or 2 units fail, however he had noticed supply water temperatures drop.
 - During the site visit a circulator pump for West building was not functional.
 - Hot water unit condition appeared reasonable, however pipework insulation was deteriorating. We note several units have been replaced over the years.
- Pool water heating systems: 2x heating water systems, 1 each for the lap pool and plunge pool. Each containing one 94kW natural-draft hot water boiler.
 - Boilers were not operational during site visit, since pools are not heated during winter months (May to September).
 - No reported operating issues with the heaters. Heaters normally require 3-5 days to warm up pools.
 - Boilers are original. Condition appeared average, although some corrosion was noted at the base of the casings. Building Manager reported recent maintenance works to replace components for one heater.
- Laundromat:
 - 8x commercial gas tumble dryers to the laundromat.
 - Dedicated gas and electric hot water units.
 - Owner was not present during site visit.

Challenges / constraints identified:

- Large-capacity gas systems (especially hot water, but also pool heating). Electric replacement options for such capacities may introduce issues with electrical supply, require larger plantroom spaces, add weights to rooftops, etc.
- Commercial tenant gas usage. Although alternative equipment options for the laundromat exist, the tenant must be willing to changeover to electric equipment. There are separate gas supplies to retail tenants and as such little to no impact on remainder of building progressing with electrification.

Notes & Minutes

- Limited available external areas for hot water and pool heating equipment (limited slab areas available on roofs).
- Additional concerns raised by residents during meeting:
 - Hot water metering and associated billing – meters are a gas company asset, how does energy and/or water get apportioned with electricity.
 - Potential increase in insurance costs for basement EV charging.
 - Existing parking issues within the laneway area.

Opportunities identified:

- Potential to reduce hot water system capacity to limit unnecessary oversizing. Base sizing on actual HW demands. Proposed to undertake logging of HW usage. This may result in a loss of redundancy meaning a fault in one unit will be a more noticeable impact to hot water temperature/supply.
- Hot water storage vs instantaneous systems. Peak usage could be met by stored hot water with smaller capacity heaters, putting less dependence on large-capacity hot water heaters to meet the peak hot water as per the existing system.
- There are some potentially spare rooftop slab areas for additional plant.

3. Requirements & Priorities:

- Requirements and priorities were presented as follows to the OC members for consideration. These will form the basis for assessing the options during the feasibility study:

Essential Requirements	
– Achieve full electrification of all gas equipment – Provide Electric Vehicle charging facilities – Comply with Territory Planning requirements and Noise Limits – Avoid significant disruptions to occupants during works – Works within Constraints of Building	
Desirable Requirements	
Low Cost	45%
– Low capital costs	20%
– Low operating costs (energy, maintenance)	20%
– Low replacement costs	5%
High Performance & Robustness	20%
– High climate adaptability	5%
– High usage adaptability	5%
– Resilient	5%
– Simple systems	5%
Low Environmental Impacts	25%
– Low aesthetic impact	5%
– Low acoustic impact	10%
– Environmentally sustainable solutions	10%
Other	10%
– Address existing deficiencies (e.g. age, condition, etc.)	5%
– Minimise disruptions to occupants	5%

- Specific comments provided by OC members regarding requirements for electrification:
 - Consider all electrical works (including running cabling) to be completed in one go rather than staged, including EV works, to limit double handling and piece-meal works.
 - Minimising disruptions to residents during works could be a higher percentage. Large and small disruptions could have different impacts. The requirements presented in the slides have been amended to the above to make avoiding significant disruptions/outages an essential requirement, whereas minimising smaller disruptions as a desirable requirement.
 - Agreement with acoustics being an important consideration. Requirements have been reworded post meeting to clarify intent. It is considered an essential requirement that new electric options do not exceed noise level limits prescribed by Access Canberra (per the <https://www.accesscanberra.act.gov.au/noise-management/noise-standards>)
 - Potential for aesthetic requirements to be removed or lowered.
 - Gas network connection charges being charged to each individual unit, where gas is not supplied to each unit (noting there is one gas line, and four gas meters, to the complex), was raised as a concern. Jacinta noted gas connection charges are generally regulated at the national level by the Australian Energy Regulator and are subject to National Gas Law and Rules, but that this was an area of interest and investigation for ACT Government as the local gas network is decommissioned..
- It was acknowledged that there will unlikely be complete agreement on priorities and relative importance across a large group of owners and residents. The intent of the above requirements and weightings is to provide a reasonable framework to assess options against. Acknowledging the above OC member comments, GHD believe the proposed requirements and weightings are considered reasonable and moving forward these will be applied during the feasibility study.

Notes & Minutes

4. Upgrade Options:

Electrification upgrades:

- GHD proposed electrification upgrade options to the OC members for the gas equipment identified. The options cover a range of technologies that have different strengths and weaknesses with respect to the requirements and existing building constraints. These options are summarised below:
- Hot Water replacement options:
 - a. Electric-resistive storage units (pros – lower capital cost, potentially reuse existing rooftop plant areas; cons – high power draw, high energy costs).
 - b. Air-source heat pumps, with storage tanks (pros – lower power draw / energy costs; cons – higher capital costs, larger outdoor plant space required, noisier, performance impacted by ambient conditions).
 - c. Air-source heat pumps with CO₂ refrigerant, and storage tanks (similar to Option 2, although environmental benefits of using a natural refrigerant).
- Pool Heating:
 - a. Electric flow boilers (pros – lower cost, reuse existing plantrooms; cons – high power draw).
 - b. Air-source heat pumps (pros – lower power draw and energy costs; cons – larger outdoor plant area required, noise risks).
- Commercial Dryers:
 - a. Like-for-like replacement with electric alternatives.
- If any additional options become available as the feasibility study progresses, GHD may consider these. Otherwise, no other options were specifically requested to be considered by the OC members and residents during the meeting.

Other upgrades:

- As part of the electrification feasibility study, GHD will also assess options for providing Electric Vehicle charging facilities and increasing the solar power capability of the site. The proposed options include:
- EV charging:
 - a. Shared facility – Shared parking bays (nominally the surface-level outdoor parking) to be provided with EV fast chargers. This will be a common service with residents charged per use via an app.
 - b. Individual per apartment – Common dedicated EV charging switchboards with metering, in which residents can then connect their own dedicated EV slow chargers at their own cost. Energy management system would be required, as a large take-up from residents would require significant electrical upgrade.
- Photovoltaic panels:
 - a. Extend existing PV provisions, to determine the maximum capacity that can be achieved within constraints and what net benefits this would provide to the owners.
 - b. Consider battery storage with the PV panels.

5. Where to from here:

- Progress the options analysis:
 - Potentially additional site visits to check feasibility and log the hot water demand profiles.
 - Complete options analyses, impacts assessment, costings, energy estimates, ROI, etc.
- Deliverable - Electrification Feasibility & Pathway Report (incl. options, costings, etc.).

☒ Attachments: GHD slide pack

This confirms and records GHD's interpretation of the discussions which occurred and our understanding reached during this meeting. Unless notified in writing within 7 days of the date issued, we will assume that this recorded interpretation or description is complete and accurate.

NOTE: If the information in this report does not agree with your record of this meeting or if there are any omissions, will you kindly advise this office immediately, otherwise we shall assume its contents to be correct.

Distribution: All Present/Absent



→ Gerry Hackett
Mechanical Engineer

Sustainable Apartments Pilot

Southport Apartments

325 Anketell St Greenway ACT

Owners Corporation Meeting

Welcome

Agenda

1. Introductions
2. Purpose of Meeting
3. Observations from Site Visits:
 - a) Summary of gas equipment
 - b) Identify challenges / constraints
 - c) Identify opportunities we have noted
4. Stakeholder Requirements:
 - a) Requirements & Priorities
 - b) Confirm any specific concerns OC members have with electrification.
5. Electrification Upgrade Options:
 - 1) Options
 - 2) General overview of technologies available, pros and cons, etc.
 - 3) Options Proposed
 - 4) Feedback

Summary of Gas Equipment

1. Domestic Hot Water:

- Stage 1 central hot water = $14 \times 45.9 \text{ kW} = \underline{642 \text{ kW}}$
- Stage 2 central hot water = $12 \times 48.8 \text{ kW} = \underline{585 \text{ kW}}$

2. Pool Heating:

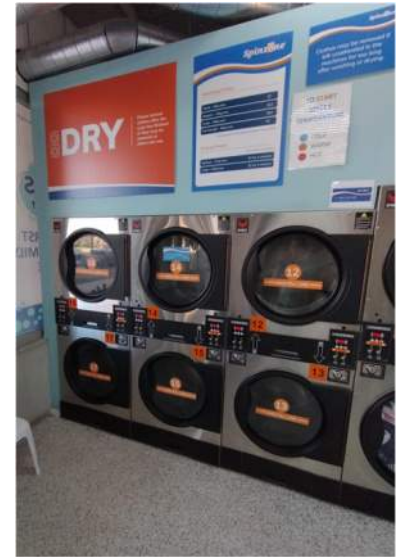
- Lap Pool = $1 \times 94 \text{ kW} = \underline{94 \text{ kW}}$
- Plunge Pool = $1 \times 94 \text{ kW} = \underline{94 \text{ kW}}$

3. Commercial Laundry

- Tumble Dryers = $6 \times 21\text{kW} + 2 \times 25\text{kW} = \underline{176 \text{ kW}}$

Other:

- 50kW photovoltaic system
- No Electric Vehicle Charging



Challenges / Constraints

1. Large-capacity gas systems
2. Commercial tenants use gas
3. Limited outdoor space (ground level, rooftop)
4. Electrical capacity (1500kVA substation)
5. Roof's ability to support weight

Any others?

Opportunities

1. DHW system – oversizing
2. DHW system – Storage vs Instantaneous

Any others?

Requirements

1. Essential Requirements
2. Desirable Requirements
3. Priorities

Are there specific concerns the OC have with electrification?

Key Category	Sub-Category	Relative Importance
Essential Requirements		
	– Achieve full electrification of the apartment building services	-
	– Work within constraints of the existing building.	-
	– Provision of PV	-
	– Provision of EV charging	-
Desirable Requirements		
Lifecycle Cost	Capital cost	20%
	Maintenance cost (incl. ease of maintenance)	5%
	Energy cost (incl. energy efficiency)	15%
	Replacement cost	5%
Robustness & Performance	Climate adaptability (e.g. extreme temperature, weather events, etc)	5%
	Usage adaptability (e.g. occupant rates, usage patterns, future expansion/reduction, etc)	5%
	Resilience & redundancy	5%
	Simplicity	5%
Environment	Aesthetics (incl. amenity, statutory planning, etc)	5%
	Acoustics	10%
	Sustainability (e.g. refrigerant emissions)	10%
Other	Address existing deficiencies (e.g. age, condition, performance)	5%
	Minimise disruptions to apartment during upgrade works.	5%

Electrification Upgrade Options

Options to represent different objectives

1. Option 1 – “Like-for-like replacement with electric-equivalent” option.
Prioritises low capital cost, ease of retrofit, minimise disruptions to apartments during upgrade works (i.e. downtimes of systems, electrical, noise, building works).
2. Option 2 – “Sustainability option” option. Prioritises low energy consumption (and energy costs), low emissions.

Other alternates:

- Worth considering a decentralised option?
- Worth considering a

Technology Options

Electrification

- Domestic Hot Water
- Pool Heating
- Commercial Dryers

Other (PV panels, batteries, EV charging)

	Pros	Cons
Electric – Instantaneous 	- Low capital cost - Easy to retrofit - Flexibility - Small footprint	- High energy costs - High electrical demand - BCA compliance challenges
Electric – Storage 	- Low capital cost - Lower electrical demand than instantaneous.	- High energy costs - Increased weights
Electric – Heat Pump 	- Low energy costs - Low electrical demand	- High capital costs - Large footprint - Install outdoors - Noisy - Increased weights - Performance impacted by ambient conditions.

Options Proposed

1. Domestic Hot Water
 - a) Option 1 – Electric instantaneous (or storage)
 - b) Option 2 – Air-source Heat pumps w/ storage
 - c) Option 3 – Air-source Heat pumps (CO2) w/ storage
2. Pool Heating
 - a) Option 1 – Electric flow boilers
 - b) Option 2 – Air-source Heat pumps
3. Commercial Dryers
 - a) Option 1 – like-for-like electric replacement

Outdoor Locations

